

Mathematics for Machine Learning

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Fourier Series

Definition

The function $T : \mathbb{R} \rightarrow \mathbb{R}$ is called a trigonometric polynomial of degree n , if there exist sequences of real numbers $\{a_k\}_{k=0}^n, \{b_k\}_{k=1}^n$ such that $a_n^2 + b_n^2 > 0$ and

$$T(x) = a_0 + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx).$$

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Theorem

If the trigonometric series $T(x) = \frac{a_0}{2} + \sum_{n=1}^{+\infty} (a_n \cos nx + b_n \sin nx)$ is uniformly convergent on $[-\pi, \pi]$, then

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} T(x) \cos kx dx, k \in \mathbb{N} \cup \{0\},$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} T(x) \sin kx dx, k \in \mathbb{N}.$$

Definition

Let f be a 2π periodic function and it is integrable on $[-\pi, \pi]$ and

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos ktdt, b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin ktdt, k \in \mathbb{N} \cup \{0\}.$$

a_k and b_k are called Fourier coefficients of f and the trigonometric series

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Fourier Series

Let $x_0 \in \mathbb{R}$ f has finite limits $f(x_0-)$ and $f(x_0+)$. Denote

$$S_0 = \frac{f(x_0-) + f(x_0+)}{2}$$

and $\varphi(t) = f(x_0 + t) + f(x_0 - t) - 2S_0$.

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Dini's Theorem

If there exists $\delta \in (0, \pi)$ such that

$$\int_0^\delta \frac{|\varphi(t)|}{t} dt < +\infty,$$

then the Fourier series of f converges to S_0 at x_0 .

Lipschitz's Theorem

If f is continuous at x_0 and for all $t \in (-\delta, \delta)$ holds

$$|f(x_0 + t) - f(x_0)| \leq L |t|^\alpha,$$

for some $\alpha \in (0, 1]$, then the Fourier series of f converges to $f(x_0)$ at x_0 .

Fejer's Theorem

Let $f \in C(\mathbb{R})$ be 2π periodic function and S_n is the n -th partial sum of the Fourier series of f , then $\sigma_n(x) \Rightarrow f(x)$, $x \in \mathbb{R}$, where

$$\sigma_n(x) = \frac{S_0(x) + \dots + S_{n-1}(x)}{n}, x \in \mathbb{R}, n \in \mathbb{N}.$$

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Complex Spaces

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Proposition

Prove that $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n x_i \overline{y_i}$ is an inner product on $\mathbb{C}^n$.

An important orthonormal basis for C^N , is the normalized Fourier basis defined as

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Discrete Fourier Transform

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The discrete Fourier transform (DFT) of the complex sequence $\{x_n\}_{n=0}^{N-1}$, is the following sequence

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The inverse discrete Fourier transform (IDFT) of the complex sequence $\{X_n\}_{n=0}^{N-1}$, is the following sequence

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Discrete Fourier Transform

$$\begin{pmatrix} \text{Diagram 1} & \text{Diagram 2} & \text{Diagram 3} & \text{Diagram 4} & \text{Diagram 5} & \text{Diagram 6} & \text{Diagram 7} & \text{Diagram 8} \\ \text{Diagram 9} & \text{Diagram 10} & \text{Diagram 11} & \text{Diagram 12} & \text{Diagram 13} & \text{Diagram 14} & \text{Diagram 15} & \text{Diagram 16} \\ \text{Diagram 17} & \text{Diagram 18} & \text{Diagram 19} & \text{Diagram 20} & \text{Diagram 21} & \text{Diagram 22} & \text{Diagram 23} & \text{Diagram 24} \\ \text{Diagram 25} & \text{Diagram 26} & \text{Diagram 27} & \text{Diagram 28} & \text{Diagram 29} & \text{Diagram 30} & \text{Diagram 31} & \text{Diagram 32} \\ \text{Diagram 33} & \text{Diagram 34} & \text{Diagram 35} & \text{Diagram 36} & \text{Diagram 37} & \text{Diagram 38} & \text{Diagram 39} & \text{Diagram 40} \\ \text{Diagram 41} & \text{Diagram 42} & \text{Diagram 43} & \text{Diagram 44} & \text{Diagram 45} & \text{Diagram 46} & \text{Diagram 47} & \text{Diagram 48} \\ \text{Diagram 49} & \text{Diagram 50} & \text{Diagram 51} & \text{Diagram 52} & \text{Diagram 53} & \text{Diagram 54} & \text{Diagram 55} & \text{Diagram 56} \\ \text{Diagram 57} & \text{Diagram 58} & \text{Diagram 59} & \text{Diagram 60} & \text{Diagram 61} & \text{Diagram 62} & \text{Diagram 63} & \text{Diagram 64} \end{pmatrix} \cdot \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{pmatrix} = \begin{pmatrix} X_0 \\ X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \\ X_7 \end{pmatrix}$$

Proposition

If $\{x_n\}_{n=0}^{N-1}$ is a sequence of real numbers and $\{X_n\}_{n=0}^{N-1}$ is its DFT, then

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$$\sum_{n=0}^{N-1} |x_n|^2 = \sum_{n=0}^{N-1} |X_n|^2$$

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This operation is called circular convolution. The convolution theorem is true for circular convolution.

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Spectrum and Phase

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